

Lesson 5

HealthCheck Utility

Completed lesson objectives

- Explain about HealthCheck tool
- Discuss the importance of HealthCheck tool
- Explain the procedure to install HealthCheck tool
- Describe the steps to customize HealthCheck tool
- Explain the importance of the scripts in Sterling B2B Integrator.

HealthCheck Tool

- HealthCheck is a utility that is packaged with IBM B2B Integrator.
- HealthCheck provides system diagnostics, error analysis, and reporting to help maintain or improve system performance.

Suggestions

- HealthCheck includes suggested actions to help system perform efficiently.
 - Provides a **simple** tool and **interactive** UI to flag common errors and potential issues allowing customers to **self-diagnose** and address **common** issues **before** they are in a production down situation.
 - **Recommends fixes** and provide information to customers explaining how to address problems.

Suggestions

(contd..)

- **Reduces** the number of Critical Situations, the number of PMRs, and the time to reach resolutions for customers since the majority of these are due to incorrect configurations.
- Establishes a **community-driven framework** for the Engineering, Services, Support, and Partners to develop ideas for scanners and contribute resolution suggestions for common issues.
- **Provides jump start** for consulting engagements between Lab Services and Partners by providing “**best practice**” implementations, allowing consultants to focus on more **customer-specific tuning**

HealthCheck Tool

- Sterling B2B Integrator v.5.2.4.2 and earlier:
 - Acquire the healthcheck jar file from the Fix Central website
 - Navigate to the Sterling B2B Integrator installation directory
 - Execute the command to stop SI (e.g., *hardstop.sh/cmd*)
 - Run the Install Service, pointing to the location of the healthcheck jar file
 - Execute the command to restart SI (e.g., *run.sh/cmd*)
- In B2Bi 5.2.5, it is integrated into the application and is installed automatically when B2Bi is installed.
- Access on browser - *<host>:<baseport>/healthcheck with dashboard user credentials.*
- The tool provides recommendations for proactive tuning and best practices.

Scanners

Scanners are collections of tests and subtests that are grouped by function. This tests ensures

optimization in multiple components of Sterling B2B Integrator.

Following are the scanner components the scanners:

- Business Process Scanner
- Config Scanner
- Database Scanner
- Dynamic Tracking Data Scanner
- Host System Scanner
- Log Scanner
- Password Scanner
- Schedule Scanner
- Static Config Scanner
- System Process Scanner

Enhancements in v1.0.3.1

- User interface and 10x.jar is updated to access Healthcheck from Internet Explorer.
- Password scanners ensures the strength of user's passwords is secured and consists of two user test, one super user test and the all user test.
- Set a limit on the number of failures a particular test can report, to avoid reworking on similar errors.
- The default scanner's testTimeout property is adjusted to 120 seconds (2 minutes) and scanner timeout property for the scanner is changed to 1200 seconds(20 minutes).
- The PDF report bar graphs use an integer scale on the y-axis. The test and groups failures in a particular test is reported based on occurrences.
- Index fragmentation, table compression and index compression test can be selected to exclude in the database scanner.

Importance

- HealthCheck tool can automate the system health checks and flag common errors and potential issues allowing customers to self-diagnose.
- It provides the recommendations to resolve any warnings or errors that are identified by the utility in the user interface.
- It provides guidelines and suggested actions to help the Sterling B2B Integrator system perform efficiently.

HealthCheck User Interface

192.168.40.100:9000/healthcheck/#properties

  Search

HealthCheck v.1.0.3.1

Tests

Tests

Add Property

Hide

Reveal

Enable

Disable

Enable Sub-Tree

Disable Sub-Tree

Expand All

Collapse All

Exp

Property Name

Property Value

Tests

Reports

Timeout

Test Error Limit

Business Process Scanner

☒ Enabled

Config Scanner

☒ Enabled

Database Scanner

☒ Enabled

Scanner Test Report

Run



-----Config Scanner:1/10-----

Config Scanner:Starting Expired Certificates Test
 Config Scanner:Finished Expired Certificates Test
 Config Scanner:Starting Nursery to Heap Test
 Config Scanner:Finished Nursery to Heap Test
 Config Scanner:Starting Thread Stealing Test
 Config Scanner:Finished Thread Stealing Test
 Config Scanner:Starting Jetty Threads Test
 Config Scanner:Finished Jetty Threads Test
 Config Scanner:Starting Customer Overrides Test
 Config Scanner:Finished Customer Overrides Test
 Config Scanner:Starting jGroups Configuration Test
 Config Scanner:Finished jGroups Configuration Test
 Config Scanner:Starting Heap High Water Mark Test
 Config Scanner:Finished Heap High Water Mark Test
 Config Scanner:Starting Multiple Ports Test
 Config Scanner:Finished Multiple Ports Test

-----Static Config Scanner:2/10-----

Static Config Scanner:Obtaining AE Execute Cycle
 information...
 Static Config Scanner:Checking queue 1
 Static Config Scanner:Calculating average cycle time...
 Static Config Scanner:Starting Average Cycle Test
 Static Config Scanner:Finished Average Cycle Test
 Static Config Scanner:Starting Cycle Time Test

✓ **Testing Complete!**

[View Summary](#)

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Status	Scanner	Results
✓	Config Scanner	3 errors
✓	Static Config Scanner	0 errors
✓	Host System Scanner	1 errors
✓	Database Scanner	14 errors
✓	Dynamic Tracking Data Scanner	0 errors
✓	Schedule Scanner	0 errors
✓	Password Scanner	37 errors
✓	System Process Scanner	0 errors
✓	Business Process Scanner	0 errors

Severity status

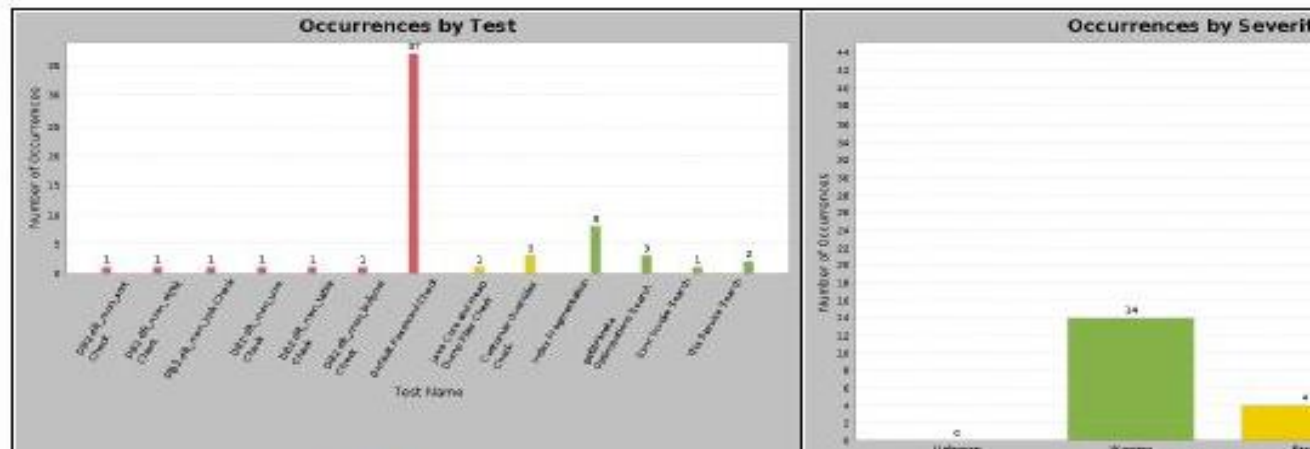
The summary report contains the severity status and color indicates as shown in the table:

Code Key	Description
1 - Warning	is in lowest level, is displayed in green in the reports and the user interface
2 - Error	is in second highest severity level, is displayed in yellow or orange.
3 - Severe	is in highest severity level, is displayed in red.
Other - Unknown	any other error, is displayed in blue

Summary Report

Summary Table

Severity	Test	# Occurrences	Suggested Action
Severe	Default Password Check	37	Change password to something more secure
Severe	DB2 dft mon bufpool Check	1	Change dft mon bufpool to ON
Severe	DB2 dft mon lock Check	1	Change dft mon lock to ON
Severe	DB2 dft mon sort Check	1	Change dft mon sort to ON
Severe	DB2 dft mon stmt Check	1	Change dft mon stmt to ON
Severe	DB2 dft mon table Check	1	Change dft mon table to ON
Severe	DB2 dft mon uow Check	1	Change dft mon uow to ON
Error	Customer Overrides Check	3	Do not modify property files directly but instead use customer overrides.properties.
Error	Java Core and Heap Dump Files Check	1	Examine and address any issues reported in the heap dump manually generated. Move or delete the file(s): heapdump.20160816.072443.12421.0005.phd, javacore.20160816.072443.12421.0006.txt
Warning	This Service Search	2	Use getDocInfo or getBPInfo instead of THIS SERVICE
Warning	Index Fragmentation	8	Consider rebuilding the index. In some cases, the reorg table reorganization, even after a table reorg operation, analyze reorgchk utility suggestions and assess the potential costs of performing a reorganization.
Warning	getBPMeta Optimizations Search	3	Set DISPOSITION and LINKAGE to false for BPMetaData; disposition and linkage data is needed.
Warning	Sync Invoke Search	1	Replace SYNC with INLINE invocation method.



Commands

- Reports can be created and viewed through the command line interface. To configure and customize tests, extensible properties can be used.

The table depicts the list of commands used to run the HealthCheck utility from the command line:

Command	Description
-h	help command lists the abbreviated scanner and test names.
-ALL	enables and runs all scanners and tests in HealthCheck.
-S	scanner command enables and runs scanners with the scanner short name. For Example, -S BPS SPS, where BPS is BusinessProcess Scanner, SPS is System Process Scanner.
-T	test command enables and runs specific test with the scanner short name. For example, -T BPS.xPath, run only XPath test in the Business Process Scanner.

Business Process Scanner Commands

Table 1. Business Process Scanner - Test Names and Properties. Displays each test in the Business Process Scanner with the command-line short name and property value.

Test Name	Command-Line Short Name	Property
BPMeta	BPS.BPMeta	BusinessProcessScanner.BPMeta
Doc Info	BPS.DocInfo	BusinessProcessScanner.DocInfo
Msg To Child	BPS.msgToChild	BusinessProcessScanner.msgToChild
Sync	BPS.sync	BusinessProcessScanner.sync
This Service	BPS.thisService	BusinessProcessScanner.thisService
X Path	BPS.xPath	BusinessProcessScanner.xPath
Doc To Dom	BPS.docToDom	BusinessProcessScanner.docToDom
Persistence	BPS.Persistence	BusinessProcessScanner.Persistence
Search	BPS.search	BusinessProcessScanner.search

Config Scanner Commands

Table 2. Config Scanner - Test Names and Properties. Displays each test in the Config Scanner with the command-line short name and property value.

Test Name	Command-Line Short Name	Property
Active Connections	CS.activeConnections	ConfigScanner.activeConnections
J Groups	CS.jGroups	ConfigScanner.jGroups
Load Balancing	CS.loadBalancing	ConfigScanner.loadBalancing
Ports	CS.ports	ConfigScanner.ports
Thread Stealing	CS.threadStealing	ConfigScanner.threadStealing
Expired Certificate	CS.expiredCertificate	ConfigScanner.expiredCertificate
Heap Water Mark	CS.heapWaterMark	ConfigScanner.heapWaterMark
Jetty Threads	CS.jettyThreads	ConfigScanner.jettyThreads
Nursery To Heap	CS.nurseryToHeap	ConfigScanner.nurseryToHeap
Overrides	CS.overrides	ConfigScanner.overrides

Customizing HealthCheck

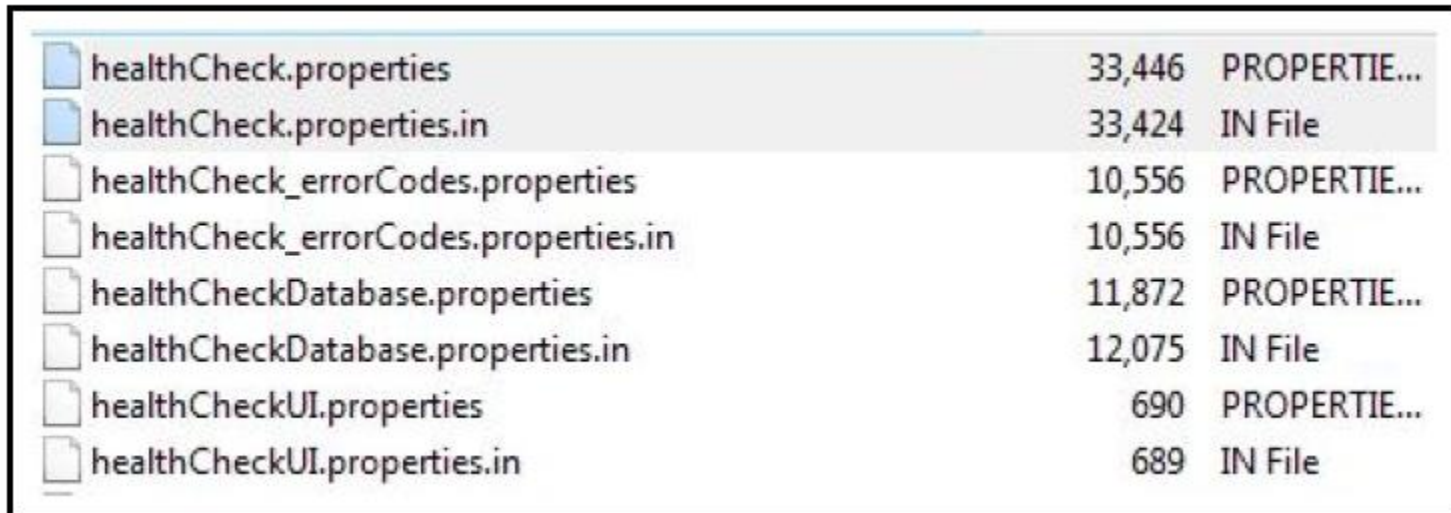
- The HealthCheck tool ensures Sterling B2B Integrator is configured as per the production or test system settings to scanner and test property settings.

Extensibility Properties

- HealthCheck includes the search and properties tests as extensibility features. The search and properties tests feature can further customized for expanded analysis through extensibility properties.
- It is not recommended to changes these settings. If the business requirement does not meet the default settings.
- You can customize property settings in two ways:
 1. HealthCheck browser user interface
 2. HealthCheck property files.

Property Files

- The HealthCheck utility tool contains all the properties in the healthcheck.properties file. The healthcheck.properties file is stored in <install_dir>/properties folder.
- Property files are depicted in the image.



A screenshot of a file explorer window showing a list of files in a directory. The files are listed with their names, sizes, and file types. The first two files, healthCheck.properties and healthCheck.properties.in, are highlighted in blue. The other files are healthCheck_errorCodes.properties, healthCheck_errorCodes.properties.in, healthCheckDatabase.properties, healthCheckDatabase.properties.in, healthCheckUI.properties, and healthCheckUI.properties.in.

healthCheck.properties	33,446	PROPERTIE...
healthCheck.properties.in	33,424	IN File
healthCheck_errorCodes.properties	10,556	PROPERTIE...
healthCheck_errorCodes.properties.in	10,556	IN File
healthCheckDatabase.properties	11,872	PROPERTIE...
healthCheckDatabase.properties.in	12,075	IN File
healthCheckUI.properties	690	PROPERTIE...
healthCheckUI.properties.in	689	IN File

Create or Edit the customer_overrides.properties file to change the properties for customizing the HealthCheck related properties.

Overview

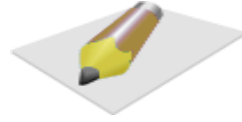
There are scripts and files created in the bin directory when the Sterling B2B Integrator is installed.

The scripts and files are in the install_dir/bin directory.



Depending on the operating system of Sterling B2B Integrator, script files have one of the following file extensions:

- .sh for UNIX
- .sh.date for UNIX
- .cmd for Windows
- .cmd.date for Windows



Exercise

1. Access HealthCheck User Interface
2. HealthCheck Command
3. Access HealthCheck Property File
4. Execute Business Process
5. Create Database Details

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Questions

